IDE 302 - P1 ORBIT COFFEE MACHINE

DESIGN MILESTONES

Beyza Dinçer S034188

MILESTONE 1: EMPHASIZE (EMPATİ)

Empathize → Define → Ideate → Prototype → Evaluate

#CoffeeRoutine #SharedCoffeeMachine #PersonalPreferences #MorningSpeed #MultiUserExperience #TokenIdentity #CollectibleTokens #CoffeeRitual #Customization

What challenges might people experience when several users share the same coffee machine but have different coffee preferences?

AI Response Summary

The AI response highlighted that shared coffee machines often create small but frequent frustrations in everyday routines.

The main insights were:

- People have different coffee preferences (strength, milk ratio, cup size).
- Adjusting the machine every time before making coffee slows down the morning routine.
- Many coffee machines are designed for generic use rather than personal habits.
- Users often want a faster and simpler interaction, especially during busy mornings.

These insights suggested that personalization and speed are important factors in shared coffee environments.

Reflection

Through this step I realized that the real issue is not only about making coffee, but about how different users interact with the same machine.

Instead of forcing users to adjust settings every time, the product could allow instant recognition of personal preferences.

This led me to think about a physical interaction system, where each user could activate their coffee profile quickly.

The idea of collectible tokens representing different users started to emerge as a potential design direction.

Magnets on the refrigerator door

#collection #biriktirmek #sergilemek #manyetik

MILESTONE 2 — DEFINE

#CoffeeRoutine #PersonalCoffeeProfile #SharedAppliance #InstantInteraction #MultiUserExperience #CoffeeRitual #TokenIdentity #Customization

User Insight

Users often share the same coffee machine but have different coffee preferences. However, most machines require users to manually adjust settings every time, which interrupts the natural flow of the morning routine. This creates small but repetitive friction in everyday coffee preparation.

AI Role: Re-framer

"How might we redesign the interaction with a shared coffee machine so that each user can access their personal coffee preferences instantly and intuitively?"

AI Response Summary

The AI suggested reframing the problem from "making coffee faster" to "making the interaction more personal and immediate."

Key ideas included:

- The machine should recognize the user quickly.
- Personal settings should be activated automatically.
- The interaction should be simple and physical rather than menu-based.
- A symbolic or tangible element could represent user identity and preference.

This reinforced the idea that personalization could be embedded into the interaction itself, rather than hidden inside digital menus.

Reflection

During this stage I realized that the real opportunity is not simply improving coffee machines, but rethinking how users interact with shared appliances. Instead of navigating settings every time, the product could allow users to activate their identity and preferences through a simple action. This insight strengthened the concept of token-based interaction, where each token represents a personal coffee profile. It also introduced the idea that the tokens could become collectible objects, adding emotional value and ritual to the coffee experience.

Design Decisions

✅ **Accepted**

- The machine should identify users instantly.
- Personal coffee settings should be linked to a physical object.
- Interaction should remain simple and intuitive.

❌ **Rejected**

- Systems that rely on complex digital interfaces or app-based setup for daily use.
- Interactions that require multiple manual adjustments each time.

🔄 **Transformed**

- The problem shifted from "how to make coffee faster" to
- "how to create a personalized interaction with a shared coffee machine."

This transformation led the concept toward a token-based identity system, where users activate their coffee preferences through their personal token.

MILESTONE 3 — IDEATE

#TokenInteraction #CollectibleDesign #UserIdentity #CoffeeCustomization #PhysicalInterface #SharedAppliance #CoffeeRitual #InteractionDesign

AI Role: Expander
"How could a coffee machine use physical tokens to represent different users and activate personalized coffee preferences?"

AI Response Summary
The AI expanded the idea by suggesting different ways physical tokens could function within the coffee machine.
Some of the explored possibilities included:

- Tokens representing individual users, each storing a personal coffee profile.
- A docking area where users place their token to activate their settings.
- Tokens designed as collectible objects, allowing users to identify with their coffee ritual.
- The interaction becoming a small daily ritual, where inserting the token prepares the machine for the user.

This step helped expand the concept beyond a simple technical feature and framed it as a meaningful user interaction.

Reflection
At this stage I explored how personalization could be expressed through physical interaction rather than digital interfaces. The idea of tokens became more interesting because they not only represent user profiles but also create a tangible relationship between the user and the machine. I also realized that making tokens collectible objects could strengthen the emotional connection to the product. Instead of being only functional elements, tokens could become part of a personal coffee ritual.

AI Role: Critique
"What potential weaknesses or usability problems could arise from using physical tokens as the main interaction method in a coffee machine?"

AI Response Summary
The AI identified several possible issues with the token concept:

- Tokens could be lost or misplaced.
- Users might forget their token when they want to make coffee.
- The docking system must be clear and intuitive to avoid confusion.
- The interaction should remain simple and fast, otherwise it could become frustrating.

These points highlighted important usability considerations for the design.

Reflection
The critique helped identify practical challenges in the concept. To address these concerns, the token system should be designed so that tokens are easy to recognize, store, and insert. The docking area should clearly indicate where and how the token should be placed. This feedback also confirmed that the token interaction must remain quick and intuitive, otherwise it could undermine the goal of simplifying the coffee routine.

ADesign Decisions

✅ **Accepted**

- Tokens will represent individual users and coffee profiles.
- The machine will include a central docking area for token interaction.
- Tokens will be designed as collectible elements that users can identify with.

❌ **Rejected**

- Hidden or complex token interactions that require multiple steps.
- Systems where tokens must be inserted deeply into the machine like a card slot.

🔄 **Transformed**

- Instead of a simple card-like interaction, the concept evolved into a visible and symbolic token docking system, where users place their token onto the machine to activate their coffee profile.

MILESTONE 4 — PROTOTYPE

#TokenDocking #ProductArchitecture #CoffeeMachineDesign #InteractionFlow #PhysicalInterface #ModularComponents #CoffeeRitual #UserExperience

AI Role: Expander
"How could the token interaction system be physically integrated into the design of a coffee machine so that it feels natural and intuitive?"

AI Response Summary
The AI suggested several ways to integrate the token system into the physical design of the coffee machine.
Key ideas included:

- Creating a dedicated docking area where users place their token.
- Making the token interaction visible and central to the product experience.
- Designing the docking system so that tokens sit naturally on the surface, rather than being inserted like a card.
- Using subtle visual cues such as light feedback to indicate when a token is active.

These suggestions helped translate the concept from an abstract interaction idea into a physical product feature.

Reflection
At this stage I focused on transforming the token concept into a clear physical interaction within the product design. I explored how the docking area could be positioned so that it becomes a central part of the machine interface. The token interaction should feel natural and visible, allowing users to immediately understand how the system works. This stage also helped me consider how other elements of the machine such as the coffee outlet, milk system, and screen interface could be organized around the token interaction.

Design Decisions

✅ **Accepted**

- A central token docking area that clearly communicates where the interaction happens.
- Tokens remain visible on the machine surface rather than being hidden.
- Subtle light feedback to indicate the active user token.

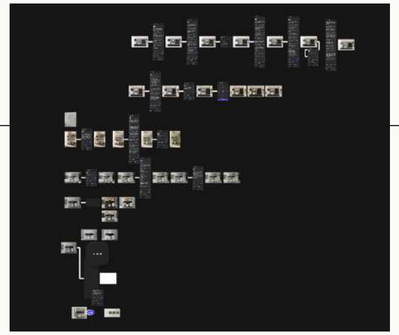
❌ **Rejected**

- Slot-based systems where tokens must be inserted like cards or keys.
- Interactions that hide the token inside the machine.
- I abandoned the idea of returning the milk from the machine through a tube from the inside for hygiene reasons.

🔄 **Transformed**

- The token system evolved from a simple identification tool into a central interaction element of the product design, shaping the layout and visual identity of the coffee machine.

MILESTONE 5 — EVALUATION & REFINEMENT



Empathize → Define → Ideate → Prototype → Evaluate

AI Role: Expander

"How could the token interaction system be physically integrated into the design of a coffee machine so that it feels natural and intuitive?"

AI Response Summary

The AI identified several potential design challenges related to the token interaction system and the product layout.

Key points included:

- Users must immediately understand where the token should be placed.
- The active token must be visually distinguishable from other tokens.
- The product layout should clearly communicate the relationship between token interaction, screen interface, and coffee output.
- The interaction must remain quick and intuitive, especially during morning routines.

These points emphasized the importance of clear visual hierarchy and interaction cues.

Reflection

During the development process, working with AI to generate product visuals and iterations revealed several limitations. Some generated images changed the proportions of the machine, added extra tokens, or modified parts of the design that were supposed to remain fixed. This created frustration and required several rounds of prompt refinement and clarification. Through this process, I realized that precise instructions and constraints are essential when collaborating with AI tools, especially in design workflows. Despite these challenges, the iterative process helped clarify the final layout of the token docking system, screen placement, and coffee flow elements.

Design Decisions

✅ **Accepted**

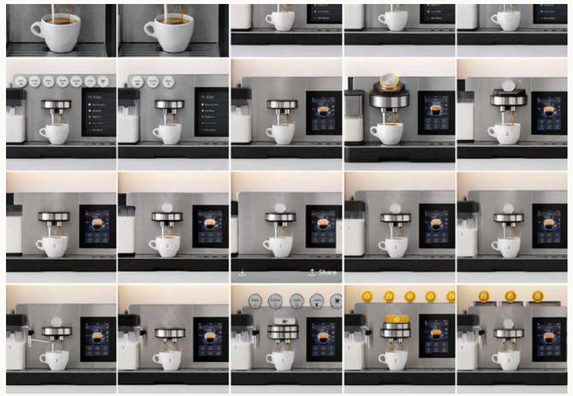
- The token docking interaction as the core element of the user experience.
- A clear visual hierarchy between the token area, screen, and coffee outlet.
- Using subtle visual feedback (lighting) to indicate the active token.

❌ **Rejected**

- Design variations where the machine proportions were unintentionally altered by AI-generated outputs.
- Layouts where the token system became visually confusing or secondary.

🔄 **Transformed**

- Early iterations treated the token simply as a technical identification feature.
- Through evaluation and iteration, the token evolved into a central visual and interaction element that defines the identity of the product.



My Recipes
Colombia Washed
18g 1:2 9 bar 93°C
Pre-infusion
Extraction
PAUSE
STOP
18.0 g
36.0 g

Alex
Emily
Ben
Alex
Hi Alex
Morning Espresso
76 mm
65 mm

Alex
Emily
Ben
Sarah
David
Linee
Lisa
Alex
Hi Alex
Morning Latte
Flat White
Cappuccino
Iced Coffee
New Recipe

Alex
Emily
Ben
Sarah
David
Lisa
Alex
Hi Alex
Morning Latte
Flat White
Cappuccino
Iced Coffee
New Recipe

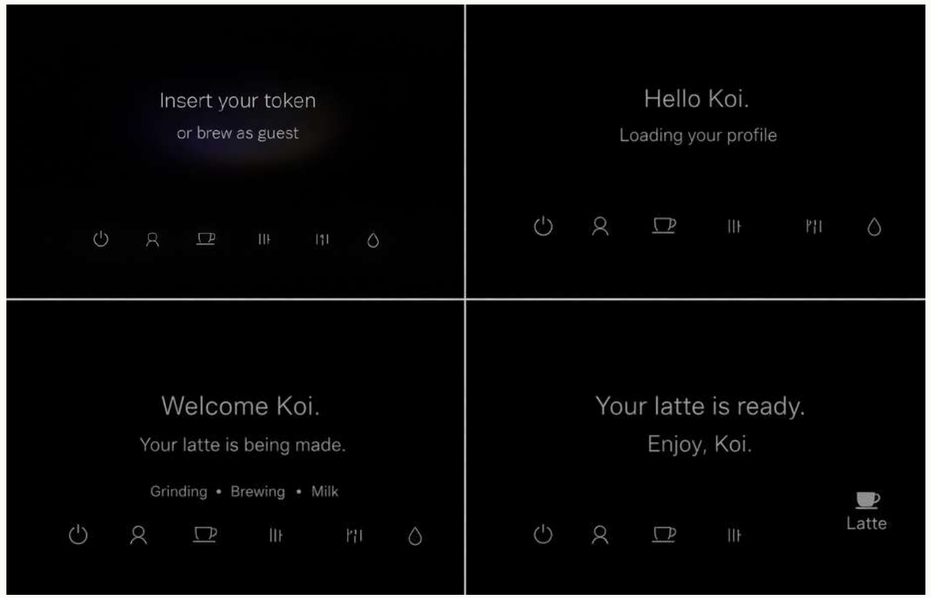
Welcome Alex
Your latte is being made

Ata
Beyza
Ege
Gülce
Zümra
Nil
Irmak
Betül
Welcome Koi.
Your latte is being made.

DETAILS

- toplam genişlik: ≈ 350 mm
- toplam yükseklik: ≈ 380 mm
- derinlik notu: ≈ 410 mm
- token çapı: ≈ 45 mm
- bardak boşluğu: ≈ 120 mm
- damlatma tepsisi: ≈ 280 mm
- süt tankı: ≈ 170 mm
- süt tankı kesiti: ≈ 90 × 90 mm

Because the tokens are magnetic, they can also be attached to surfaces such as refrigerators.



“Users can create, name, and store their own coffee recipes, turning the machine into a personal brewing library.”

Token → Profile → Recipe → Edit → Name → Brew

TOKEN → identity
DISPLAY → knowledge
DOCK → control